VBR-Net: Vessel-Guided Boundary Residual Networks for Dermatological Vessel Segmentation

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ABSTRACT

Accurate segmentation of blood vessels in dermatological optical coherence tomography angiography (OCTA) images is essential for quantitative analysis of skin microvasculature but remains challenging due to the complex morphology of vessels and pervasive speckle noise. Conventional encoder-decoder architectures often blur fine vessel boundaries and propagate background noise, limiting segmentation precision. To address these limitations, we propose Vessel-Guided Boundary Residual Network (VBR-Net), a novel architecture that incorporates explicit geometric priors to guide feature refinement. The core innovation lies in two synergistic modules: a Gradient-Orthogonal Boundary Residual Block that refines uncertain edge regions using gradient-based spatial attention, and a Centerline-Conditioned Spatial Gating Decoder that modulates skip-connection features with an estimated vessel centerline map to preserve structural continuity and suppress noise. This dual mechanism enforces geometric consistency across multiple resolutions, simultaneously recovering sharp boundaries and maintaining topological integrity. Validated on the publicly available DERMA-OCTA dataset, VBR-Net achieves superior performance in segmenting vessels with a broad spectrum of calibers, significantly outperforming baseline methods in boundary accuracy and robustness to speckle artifacts. The code and checkpoints are available at https://github.com/yifangao112/VBR-Net

1. Introduction

Optical Coherence Tomography Angiography (OCTA) has emerged as a pivotal non-invasive imaging modality for visualizing the microvasculature in dermatology Li, Murthy, Zhu, Zhang, Tang, Mehrabi, Kelly, and Chen (2021). It enables the detailed assessment of vessel morphology associated with various skin pathologies, including chronic venous disease and skin cancers, supporting diagnosis and treatment monitoring Zugaj, Chenet, Petit, Vaglio, Pascual, Piketty, and Bourdès (2018). The quantitative analysis of these vascular networks—measuring metrics such as vessel density, diameter, and tortuosity—fundamentally relies on the accurate pixel-wise segmentation of blood vessels from the acquired en face projections.

However, the automatic segmentation of dermatological OCTA vessels presents a formidable challenge due to the complex and heterogeneous nature of the imagery. The vasculature exhibits a broad spectrum of calibers, from thick trunk vessels to thin, fragile capillaries, all embedded within a speckle-corrupted background inherent to OCTA acquisition Chen, Yang, Na, and Wang (2025). This complexity necessitates a segmentation model capable of simultaneously recovering sharp, precise boundaries for vessels of all sizes while robustly suppressing noise-induced false detections. Conventional encoder-decoder architectures, such as the widely adopted U-Net Ronneberger, Fischer, and Brox (2015) and its automated variant nnUNet Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020), have become the de facto standard for medical image segmentation. These models propagate high-level semantic features to the decoder via simple bilinear upsampling or transposed convolutions. While effective for capturing global context, this process often blurs fine vessel edges and indiscriminately mixes signals from vessel interiors with background noise at boundary regions, leading to topological errors and boundary inaccuracies.

Recent advancements have sought to address these limitations. Frequency-domain approaches and multi-scale feature fusion strategies, such as those in UNet++ Zhou, Siddiquee, Tajbakhsh, and Liang (2018) and DeepLabv3+ Chen, Zhu, Papandreou, Schroff, and Adam (2018), improve feature representation but

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often fail to explicitly model the intrinsic geometric relationship between vessel centerlines and their surrounding boundary gradients. This leaves the delineation of thin-vessel boundaries, where signal-to-noise ratio is lowest, particularly under-constrained. Other works have incorporated attention mechanisms Badrinarayanan, Kendall, and Cipolla (2015) or transformer blocks Chen, Lu, Yu, Luo, Adeli, Wang, Lu, Yuille, and Zhou (2021) to enhance long-range dependencies, yet they may not provide the explicit spatial guidance needed for pixel-perfect boundary localization in the presence of severe speckle noise.

To bridge this gap, we propose the Vessel-Guided Boundary Residual Network (VBR-Net), a novel encoder-decoder architecture designed for boundary-aware segmentation of OCTA blood vessels. Our core innovation is the synergistic integration of two complementary geometric guidance modules at each decoder stage: a Gradient-Orthogonal Boundary Residual Block and a Centerline-Conditioned Spatial Gating Decoder. The boundary residual block actively identifies and refines uncertain edge regions using a gradient-based spatial attention mechanism, directly sharpening feature responses at vessel boundaries. Concurrently, the centerline gating decoder introduces a structural prior by estimating a vessel centerline probability map from intermediate features and using it to modulate the skip-connection flow. This selectively emphasizes encoder details aligned with the vessel skeleton while suppressing irrelevant background noise, thereby preserving structural continuity and mitigating false positives.

Built upon an enhanced nnUNet-style backbone with increased channel capacity, VBR-Net enforces both geometric precision and topological consistency across multiple spatial resolutions. We rigorously evaluate our method on the publicly available DERMA-OCTA dataset, comprising 331 2D projection images. VBR-Net achieves a Dice score of 0.8381 and a Hausdorff Distance (95%) of 3.0712, outperforming a comprehensive suite of state-of-the-art baselines including UMamba, UNet++, and UTNet. Ablation studies confirm the contribution of each proposed component, and subgroup analysis shows the model's particular robustness in segmenting capillary-dominant regions where boundary precision is most challenging. This work makes three primary contributions. First, we introduce a novel dual-guidance framework that explicitly incorporates vessel-specific geometric priors—boundary gradients and centerline maps—into the segmentation network architecture. Second, we design two novel modules: the Gradient-Orthogonal Boundary Residual Block for edge-aware feature refinement and the Centerline-Conditioned Spatial Gating Decoder for structure-preserving feature fusion. Third, we achieve state-of-the-art performance on a public dermatological OCTA dataset, with significant improvements in boundary accuracy, validated through extensive quantitative and qualitative analyses.

2. Related Work

2.1. Deep Learning for Vessel Segmentation

The foundational work in medical image segmentation established the encoder-decoder paradigm, with the Fully Convolutional Network (FCN) Shelhamer, Long, and Darrell (2014) demonstrating that dense pixel-wise predictions could be learned end-to-end. This architecture was significantly advanced by the U-Net Ronneberger et al. (2015), which introduced symmetric skip connections to propagate high-resolution spatial features from the encoder to the decoder, enabling precise localization. For vessel segmentation, U-Net and its variants, such as the Attention U-Net Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018), became the de facto standard, effectively capturing the tubular structures of vasculature. The nnU-Net framework Isensee et al. (2020) later systematized this approach by automating architecture and training hyperparameters, establishing a robust and highly competitive baseline across numerous medical segmentation tasks, including vessel delineation. However, these models rely on simple upsampling operations like bilinear interpolation or transposed convolutions to recover spatial resolution, a process that inherently smooths fine details and can blur the boundaries of thin capillaries, a critical limitation for high-resolution OCTA analysis.

To address the specific challenges of vascular segmentation, including topological preservation and boundary ambiguity, subsequent research has introduced specialized architectural modifications and loss functions. In contrast to standard U-Net, which treats all pixels equally, approaches like the CE-Net Gu, Cheng, Fu, Zhou, Hao, Zhao, Zhang, Gao, and Liu (2019) incorporated dense atrous convolutions to capture multi-scale contextual information without sacrificing resolution, improving the connectivity of segmented vessel trees. Other works explicitly targeted boundary errors by integrating auxiliary supervision; for instance,

the method proposed by Liu, Tang, Guo, Qin, Jia, Zhan, Zhou, and Wu (2019) utilized a Dice loss to handle class imbalance, while more recent techniques like the Boundary Loss Kervadec, Bouchtiba, Desrosiers, Granger, Dolz, and Ayed (2018) formulate a distance-based term that directly penalizes errors at the segmentation contour. Topology-aware losses have also been proposed to maintain the correct connectivity of vascular networks, addressing a common failure mode where segments are incorrectly broken or merged Ozcelik, Unver, Gurses, Turkay, and Demir (2023). Despite these advances, a core limitation persists: these methods often treat boundary refinement and structural continuity as separate, post-hoc objectives rather than as an integrated, geometrically consistent constraint enforced throughout the feature decoding process.

The pursuit of sharper boundaries and noise robustness has also led to explorations beyond the spatial domain and standard convolutional blocks. Frequency-domain approaches, such as those employing wavelet transforms or Fourier-based operations, aim to separate signal from noise in the spectral domain Zhao, Dong, Yu, Zhao, Du, and Xu (2024). While effective for certain types of structured artifact, these methods generally lack an explicit mechanism to model the spatial relationship between a vessel's centerline and its boundary gradients, which is crucial for delineating thin structures in speckle-corrupted OCTA. Concurrently, the rise of transformer architectures has introduced new paradigms for capturing long-range dependencies. Vision Transformers (ViTs) and hybrid models like TransUNet Chen et al. (2021) and SwinUNETR He, Nath, Yang, Tang, Myronenko, and Xu (2023) use self-attention to aggregate global context, showing promise for capturing extensive vascular networks. More recently, state-space models (SSMs) like Mamba Gu and Dao (2023) and its medical imaging adaptation UMamba Ma, Li, and Wang (2024) offer efficient long-sequence modeling. However, as our comparative results indicate, while these advanced architectures achieve strong overall Dice scores, they can still underperform in boundary precision metrics like the 95% Hausdorff Distance, as they do not explicitly reinforce geometric priors during feature decoding. This gap motivates the design of architectures that directly inject geometric guidance, such as vessel centerlines and boundary gradients, into the core decoding pathway to achieve simultaneous structural continuity and boundary sharpness.

2.2. Geometric and Boundary-Aware Refinement

A prominent line of research addresses boundary refinement through specialized loss functions that directly penalize contour errors. These methods typically compute an additional term that measures the discrepancy between predicted and ground truth boundaries, complementing region-based losses like Dice. The Boundary Loss Kervadec et al. (2018) formulates this as a distance map-based penalty, effectively pushing predictions toward correct edges. Similarly, Boundary IoU Valverde and Tohka (2021) introduces a metric that focuses evaluation on a narrow band around object contours, which can also be adapted as a differentiable loss. Other works incorporate gradient information directly into the loss, such as by penalizing differences in image gradient magnitudes between predictions and labels Guo, Chen, Chen, Huang, Xu, Cao, Zhao, and Tan (2018) or using a multi-scale edge-aware loss Zheng, Huan, Xia, and Gong (2020). While effective at improving contour fidelity, these loss-based approaches operate at the optimization level and do not explicitly modify the network's feature representations or architectural flow, potentially limiting their capacity to correct boundary errors arising from intermediate feature degradation.

Another category of methods integrates boundary awareness directly into the network architecture through attention mechanisms or dedicated refinement branches. Spatial attention gates, as popularized by Attention U-Net Oktay et al. (2018), dynamically highlight salient regions in skip connections, which can implicitly prioritize boundary-relevant features. More explicit approaches incorporate gradient guidance by computing image gradients as an auxiliary input or using them to modulate feature learning. For instance, some architectures employ a parallel boundary detection branch that shares features with the main segmentation stream, enforcing consistency between region and edge predictions Qin, Lin, Gao, Han, Wang, Qin, Li, An, and Che (2025). Others use learned spatial gates conditioned on gradient maps to refine features in edge regions Liu, Xia, Feng, and Bai (2023). The DeepLab series Chen, Papandreou, Kokkinos, Murphy, and Yuille (2016) employs atrous spatial pyramid pooling to capture multi-scale context, which aids in resolving boundary ambiguity. However, these mechanisms often treat boundary refinement as a separate or auxiliary task rather than an orthogonal feature modulation process that works in concert with structural priors.

A distinct strategy uses explicit geometric or topological priors to enforce structural consistency, which indirectly improves boundary placement by preserving object shape. Shape priors, often incorporated via

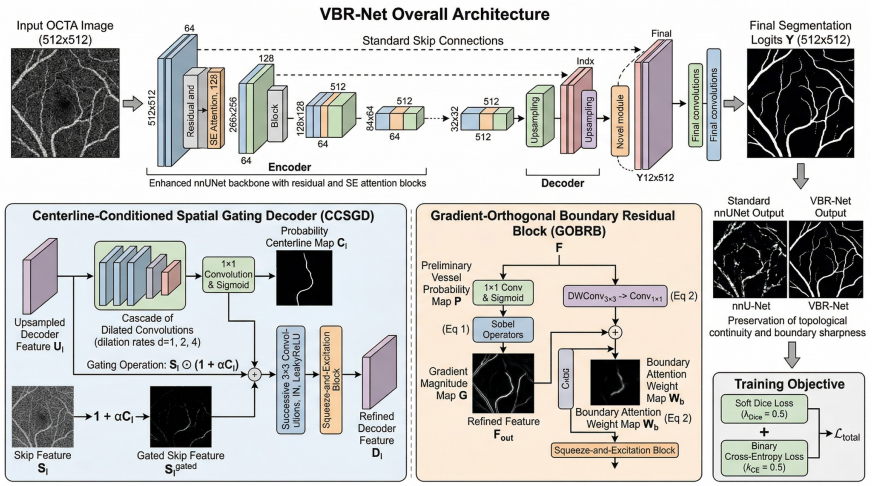

Figure 1: Overall architecture of the proposed VBR-Net. The symmetric encoder-decoder network integrates a Centerline-Conditioned Spatial Gating Decoder and a Gradient-Orthogonal Boundary Residual Block at each decoder stage to enforce geometric consistency for precise vessel segmentation.

statistical shape models or level-set formulations, constrain the segmentation to plausible anatomical forms Li, Zhang, and He (2020). For tubular structures like vessels, centerline constraints have proven particularly effective. Methods may jointly segment the vessel interior and its centerline, using the latter as a topological guide Shi, Hu, Yang, Gao, Liu, and Ma (2024), or employ a multi-task learning framework that regularizes the segmentation with a centerline distance map Li, Ma, Cheng, Paetzold, Rueckert, and Kainz (2025). The use of connectivity-preserving losses Li et al. (2025) or Euler characteristic transforms Röell and Rieck (2023) further ensures topological correctness. While these approaches excel at maintaining global structure and continuity, they frequently lack a dedicated, synergistic mechanism to simultaneously sharpen local boundaries, especially in noisy, high-resolution imagery where edge signals are weak. This creates a gap for architectures that can jointly enforce centerline-conditioned feature modulation for structural preservation and gradient-orthogonal boundary refinement for geometric precision within a unified decoding framework.

3. Method

3.1. Overall Architecture

The accurate segmentation of dermatological OCTA vasculature presents a unique challenge due to the coexistence of thick trunk vessels and thin, fragile capillaries within a speckle-corrupted background. Standard encoder-decoder architectures, while effective for many segmentation tasks, often propagate features through simple upsampling operations that indiscriminately mix semantic information, leading to blurred vessel boundaries and the propagation of background noise into the decoding pathway Liu, Shen, Yang, bin Bian, and Yu (2023). This is particularly detrimental for thin capillaries, where precise boundary localization is critical for accurate morphological assessment. To overcome these limitations, we propose the Vessel-Guided Boundary Residual Network (VBR-Net), an architecture designed to explicitly enforce geometric consistency throughout the decoding process. The core innovation of VBR-Net is the synergistic integration of two novel modules—the Centerline-Conditioned Spatial Gating Decoder and the Gradient-Orthogonal Boundary Residual Block—into an enhanced nnUNet-style backbone. This design is motivated by the intuition that vessel segmentation can be significantly improved by simultaneously leveraging a structural prior (the vessel centerline) to guide feature selection and a local geometric cue (the boundary gradient) to refine edge delineation, thereby addressing both topological continuity and boundary sharpness in a unified framework.

The overall architecture of VBR-Net, illustrated in Figure 1, follows a symmetric encoder-decoder design with skip connections, processing input OCTA images of size 512×512. The encoder is constructed as an enhanced version of the nnUNet backbone Isensee et al. (2020), employing residual blocks with

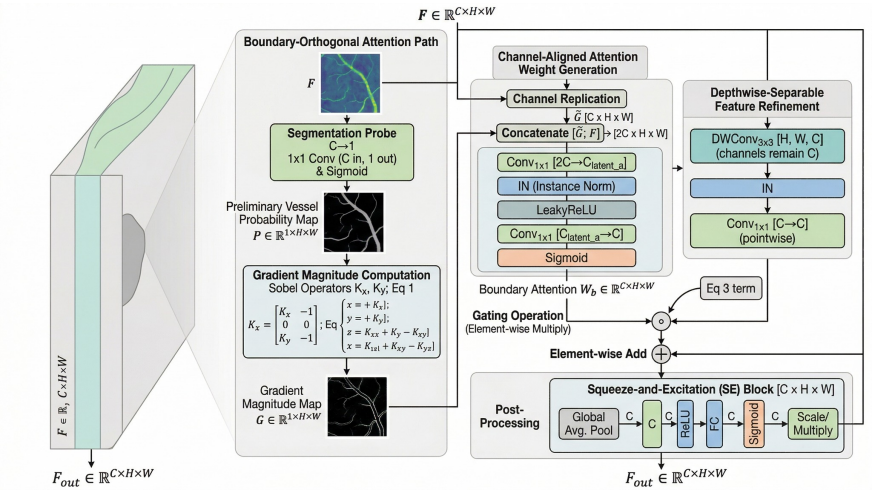

Figure 2: Detailed schematic of the Gradient-Orthogonal Boundary Residual Block. The module computes a boundary attention map from the gradient of a soft segmentation probe and uses it to weight a depthwise-separable residual refinement, sharpening features at uncertain edge regions.

integrated Squeeze-and-Excitation (SE) attention Hu, Shen, Albanie, Sun, and Wu (2017) to improve feature discriminability. It consists of four downsampling stages, each comprising an initial 3×3 convolution with instance normalization and LeakyReLU activation, followed by two sequential ResidualConvBlocks. Each block performs two 3×3 convolutions with a residual connection, and concludes with an SE block that recalibrates channel-wise feature responses. The encoder progressively extracts multi-scale features at spatial resolutions of 256×256, 128×128, 64×64, and 32×32, with corresponding channel depths of $[64, 128, 256, 512]$. This increased channel capacity relative to standard designs ensures sufficient representational power to capture the broad spectrum of vessel calibers present in OCTA imagery.

The decoder pathway is where VBR-Net introduces its novel geometric guidance mechanisms. At each decoder level l, the data flow follows a carefully orchestrated sequence designed to first establish structural coherence and then refine geometric precision. The process begins with the upsampling of the deeper decoder feature map $\mathbf{D}_{l+1}$ via a transposed convolution, producing the upsampled feature $\mathbf{U}_l \in \mathbb{R}^{C_l\times H_l\times W_l}$. This feature, along with the corresponding encoder skip feature $\mathbf{S}_l$ from the same spatial resolution, is first processed by the Centerline-Conditioned Spatial Gating Decoder. This module estimates a probabilistic vessel centerline map $\mathbf{C}_l$ from $\mathbf{U}_l$ and uses it to spatially gate $\mathbf{S}_l$, selectively increasing feature responses aligned with the estimated vessel skeleton while suppressing irrelevant background noise. The gated skip feature is then fused with $\mathbf{U}_l$ and processed to produce an intermediate refined feature $\mathbf{D}_l$. Subsequently, the Gradient-Orthogonal Boundary Residual Block is applied directly to $\mathbf{D}_l$. This block computes a spatial attention map derived from the gradient of a soft segmentation probe, identifying regions of high boundary uncertainty. It then applies a depthwise-separable residual refinement, weighted by this attention map, to sharpen feature responses precisely at these edge regions. This two-stage, per-level refinement—first centerline-guided structural filtering, followed by gradient-driven boundary sharpening—is applied consistently across all decoder resolutions, ensuring that both global topology and local geometry are reinforced throughout the reconstruction of the segmentation map.

The final decoder output at the full 512×512 resolution undergoes a final refinement stage consisting of a 3×3 convolution, instance normalization, LeakyReLU activation, and a 1×1 convolution to generate the final segmentation logits $\mathbf{Y} \in \mathbb{R}^{H\times W}$. The entire network is trained end-to-end using a combined loss function that balances region overlap and pixel-wise accuracy. The synergistic operation of the two core modules enables VBR-Net to address the fundamental challenges of OCTA vessel segmentation: the Centerline-Conditioned Spatial Gating Decoder provides a robust structural prior that mitigates false positives from speckle noise, while the Gradient-Orthogonal Boundary Residual Block delivers the localized refinement necessary for recovering sharp, accurate boundaries for both large trunks and thin capillaries.

3.2. Gradient-Orthogonal Boundary Residual Block

The Gradient-Orthogonal Boundary Residual Block is designed to counteract the progressive blurring of vessel boundaries that occurs during the upsampling stages of conventional encoder-decoder networks. In OCTA images, where thin capillaries exhibit subtle intensity transitions against a noisy background, standard bilinear upsampling or transposed convolution operations indiscriminately smooth features, leading to inaccurate boundary delineation and topological errors. This module directly addresses this limitation by generating a spatially selective residual that amplifies feature responses specifically in regions of high geometric uncertainty, thereby preserving edge sharpness across all vessel calibers.

The core intuition is to use the spatial gradient of an intermediate soft segmentation map as a proxy for boundary uncertainty. Given an input feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ from the decoder, the process begins with a lightweight segmentation probe. This probe consists of a single 1×1 convolution followed by a sigmoid activation, producing a preliminary vessel probability map $\mathbf{P} \in \mathbb{R}^{1\times H\times W}$. To quantify spatial uncertainty, we compute a gradient magnitude map $\mathbf{G}$ by applying fixed Sobel operators. This operation identifies pixels where the probability transitions rapidly, which predominantly correspond to vessel boundaries.

$$\mathbf{G}_{i,j} = \sqrt{(\mathbf{K}_x\mathbf{P})^2_{i,j} + (\mathbf{K}_y\mathbf{P})^2_{i,j} + \epsilon} \tag{1}$$

Here, $\mathbf{K}_x$ and $\mathbf{K}_y$ are the horizontal and vertical Sobel derivative kernels, $*$ denotes 2D convolution, and ϵ is a small constant for numerical stability. The resulting map $\mathbf{G}$ serves as a geometric prior, highlighting regions where feature refinement is most critical.

This gradient prior is then integrated with the original features to compute a boundary attention weight map. The gradient map $\mathbf{G}$ is first replicated across the channel dimension to yield $\tilde{\mathbf{G}} \in \mathbb{R}^{C\times H\times W}$. It is then concatenated with the input feature map $\mathbf{F}$ and processed by a compact subnetwork to generate channel-aligned attention weights.

$$\mathbf{W}_b = \sigma\Big(\mathrm{Conv}_{1\times1}\Big(\mathrm{LeakyReLU}\Big(\mathrm{IN}\Big(\mathrm{Conv}_{1\times1}\Big([\tilde{\mathbf{G}};\mathbf{F}]\Big)\Big)\Big)\Big)\Big) \tag{2}$$

In this formulation, IN denotes instance normalization and σ is the sigmoid function, ensuring the weights $\mathbf{W}_b \in \mathbb{R}^{C\times H\times W}$ are bounded between 0 and 1. These weights modulate a residual correction derived from the input features. The final refined output is produced via a residual connection, where the correction is a depthwise-separable transformation of $\mathbf{F}$, weighted element-wise by $\mathbf{W}_b$.

$$\mathbf{F}_{\text{out}} = \mathbf{F} + \mathbf{W}_b \odot \mathrm{IN}(\mathrm{Conv}_{1\times1}(\mathrm{DWConv}_{3\times3}(\mathbf{F}))) \tag{3}$$

The use of a depthwise separable convolution $\mathrm{DWConv}_{3\times3}$ followed by a pointwise convolution ensures efficient, spatially-aware feature transformation. The output $\mathbf{F}_{\text{out}}$ is subsequently passed through a Squeeze-and-Excitation block for channel-wise recalibration, further enhancing discriminative power.

This block is strategically placed after the feature fusion operation at each decoder upsampling stage. Its position is critical; it operates on the fused feature representation produced by the preceding Centerline-Conditioned Spatial Gating Decoder. This sequential processing ensures that structural continuity is first reinforced by the centerline gate, after which the boundary residual block applies localized, gradient-guided refinement to sharpen the resulting edges. By selectively increasing feature responses only at uncertain boundary zones, the module prevents the smoothing of thin capillaries while leaving confident interior and background regions largely unchanged, leading to the precise segmentation boundaries shown in Figure 4.

3.3. Centerline-Conditioned Spatial Gating Decoder

The Centerline-Conditioned Spatial Gating Decoder addresses the critical challenge of preserving the structural continuity of vessel networks while suppressing background noise in OCTA images. Standard skip connections in encoder-decoder networks propagate features indiscriminately, mixing high-frequency vessel details with speckle-induced false activations from the background. This noise propagation is particularly detrimental for thin capillaries, where weak signals can be easily overwhelmed. To mitigate this, we introduce an explicit geometric prior by estimating a vessel centerline confidence map, which serves as a spatial gate to modulate the flow of encoder features before fusion, thereby emphasizing structurally relevant information.

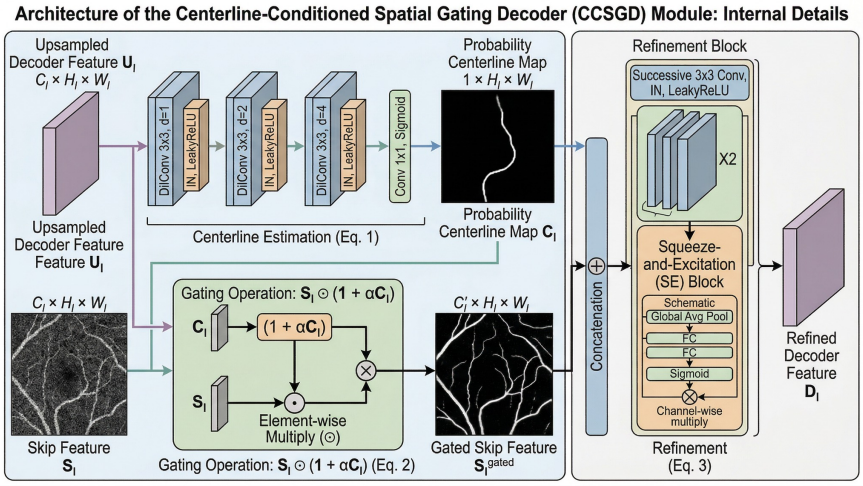

Figure 3: Architecture of the Centerline-Conditioned Spatial Gating Decoder. The module estimates a vessel centerline confidence map via a dilated convolution cascade and uses it to spatially gate encoder skip features, emphasizing structurally relevant information while suppressing background noise.

The module's design is based on the observation that vessel centerlines provide a robust topological skeleton that is less susceptible to boundary noise and intensity variations. At each decoder level l, we first obtain the upsampled decoder feature $\mathbf{U}_l \in \mathbb{R}^{C_l \times H_l \times W_l}$. A centerline confidence map $\mathbf{C}_l$ is estimated directly from this feature to capture the evolving structural understanding of the network. To effectively model vessels of varying calibers—from thick trunks to thin capillaries—we employ a cascade of dilated convolutions. This cascade progressively aggregates multi-scale contextual information, enabling the detection of elongated tubular structures across different receptive fields.

Formally, the centerline map is generated through a dedicated pathway:

$$\mathbf{C}_l = \sigma(\mathrm{Conv}_{1\times1}(\mathrm{DilConv}_{3\times3,d=4}(\mathrm{LeakyReLU}(\mathrm{IN}(\mathrm{DilConv}_{3\times3,d=2}(\mathrm{LeakyReLU}(\mathrm{IN}(\mathrm{DilConv}_{3\times3,d=1}(\mathbf{U}_l)))))))))\,, \tag{4}$$

where $\mathrm{DilConv}_{3\times3,d=r}$ denotes a 3×3 convolution with dilation rate r, IN is instance normalization, and σ is the sigmoid activation. The dilation rates of 1, 2, and 4 are chosen to capture local vessel continuity and broader connectivity patterns without excessively increasing parameter count. The resulting map $\mathbf{C}_l \in \mathbb{R}^{1\times H_l \times W_l}$ encodes the probability of each spatial location belonging to a vessel centerline.

This estimated centerline map is then used to condition the corresponding encoder skip feature $\mathbf{S}_l$ via a gating mechanism. The gating operation is designed to selectively increase feature responses aligned with the predicted vessel skeleton while attenuating contributions from off-centerline regions likely to be background noise. The gated skip feature is computed as:

$$\mathbf{S}_l^{\mathrm{gated}} = \mathbf{S}_l \odot (\mathbf{1} + \alpha \cdot \mathrm{Expand}_C(\mathbf{C}_l))\,, \tag{5}$$

where $\odot$ denotes element-wise multiplication, Expand_C replicates the single-channel $\mathbf{C}_l$ across the C_l feature channels, and α is a learnable scalar parameter initialized to 0.15. The use of $\mathbf{1}+\alpha\cdot\mathbf{C}_l$ ensures a residual-style modulation where the gate provides a positive boost to centerline-aligned features, preventing the complete suppression of information. The gated feature $\mathbf{S}_l^{\mathrm{gated}}$ and the decoder feature $\mathbf{U}_l$ are then concatenated and processed through two successive 3×3 convolutions with instance normalization and LeakyReLU activations, followed by a Squeeze-and-Excitation block for channel-wise recalibration, to produce the refined decoder feature $\mathbf{D}_l$. This integrated process ensures that topological continuity is reinforced at each decoding stage, directly suppressing false positives from speckle artifacts and preserving the connectivity of thin vascular networks.

4. Experiments

4.1. Datasets and Implementation Details

The experiments were conducted on the publicly available *DERMA-OCTA* dataset, a comprehensive collection of dermatological optical coherence tomography angiography (OCTA) scans designed for vessel segmentation analysis. This dataset comprises 331 two-dimensional average intensity projection (AIP) images derived from 330 volumetric scans of 74 subjects, including healthy individuals and patients with chronic venous disease and various skin lesions. Each image has dimensions of 512×483 pixels, corresponding to a physical scan area of 10.0×9.5 mm. The ground truth binary vessel masks were generated through a rigorous semi-automatic process using Amira software, involving four annotators and subsequent verification by a technician and a clinician to ensure reliability. Accurate segmentation of this microvasculature is clinically significant as it enables the quantitative assessment of vessel density, diameter, and tortuosity, which are crucial biomarkers for non-invasive diagnosis and treatment monitoring of skin pathologies. For our experiments, the dataset was partitioned into training and test sets using an 8:2 ratio, resulting in 265 images for training and 66 for testing.

To quantitatively evaluate segmentation performance, we employ two primary metrics: the Dice Similarity Coefficient (DSC) and the 95th percentile of the Hausdorff Distance (HD95). The Dice Similarity Coefficient measures the volumetric overlap between the predicted segmentation P and the ground truth G, defined as:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{6}$$

A higher DSC indicates better overall region agreement. For boundary precision, we use the HD95 metric, which computes the 95th percentile of the maximum distance between the boundaries of P and G, making it robust to outliers. This metric is particularly sensitive to the accurate delineation of thin capillary boundaries, where even small errors can significantly impact clinical measurements. In clinical applications such as treatment planning, an HD95 value below 4 mm is often considered acceptable for reliable quantitative analysis, underscoring the importance of this metric for assessing the practical utility of a segmentation method.

Our proposed VBR-Net was implemented within the nnU-Net framework using its standard 2D configuration. The model was trained for 100 epochs with a batch size of 12 using stochastic gradient descent (SGD) with Nesterov momentum of 0.99 and a weight decay of 3×10^{-5}. The initial learning rate was set to 0.01 and decayed following a polynomial schedule. The loss function combined Dice and cross-entropy terms. All input images were preprocessed using the standard nnU-Net pipeline, which includes intensity normalization and resampling to an isotropic resolution, and the same postprocessing steps were applied. Data augmentation followed the nnU-Net default strategy, incorporating random rotations, scaling, elastic deformations, and intensity perturbations. Training was performed on a single NVIDIA GeForce RTX 4090 GPU. For a fair comparison, all baseline models were evaluated under consistent conditions. Non-nnU-Net baselines were trained using their original architectural configurations and authors' recommended hyperparameters where available, while all methods shared the identical nnU-Net preprocessing, postprocessing pipeline, and the same 8:2 train/test split to ensure comparability.

4.2. Ablation Studies

We conduct comprehensive ablation studies to validate the contribution of each architectural component and to justify key design decisions. The experiments are performed on the DERMA-OCTA validation set, with results reported using Dice score and 95th percentile Hausdorff Distance (HD95).

4.2.1. Component Removal Analysis

To quantify the contribution of each proposed module, we systematically remove them from the full VBR-Net architecture. Table 1 presents the results. The full model, integrating all components, achieves the best boundary precision with an HD95 of 3.0670. Removing the Gradient-Orthogonal Boundary Residual Block (BR) leads to a measurable degradation in boundary accuracy, increasing HD95 by 0.0215. This confirms the block's role in refining uncertain edge regions and sharpening vessel contours. Removing the

Table 1
Component removal ablation results.

Configuration	Removed Component	Dice Score	HD95 Score
Full Model	-	**0.8397**	**3.0670**
w/o BR	Boundary Refinement	0.8397	3.0885
w/o CG	Centerline Gating	0.8415	3.1464
w/o SE	Squeeze-and-Excitation	0.8386	3.1733

Table 2
Comparison of boundary weight computation (BWC) strategies.

Variant	HD95 Score
Proposed (gradient_only)	**3.0670**
concat	3.2988
additive	3.2144
multiplicative	3.1857
feature_only	3.4238

Centerline-Conditioned Spatial Gating Decoder (CG) results in a more substantial increase in HD95 (0.0794), highlighting its critical function in suppressing background noise and preserving structural continuity by gating skip connections with a centerline prior. While the Dice score slightly increases in this configuration, the significant rise in boundary error indicates a trade-off where the model may produce more contiguous but less precise segmentations. Removing the Squeeze-and-Excitation (SE) attention modules from the residual blocks also degrades performance, particularly on the HD95 metric, underscoring the importance of channel-wise feature recalibration for focusing on salient vascular structures. The collective degradation observed from removing any single module validates the synergistic design of VBR-Net.

4.2.2. Boundary Weight Computation Strategy

The Gradient-Orthogonal Boundary Residual Block computes a spatial attention map to weight its residual correction. We investigate several alternative mechanisms for generating this boundary weight map. Table 2 compares the proposed method against four variants. The ‘concat‘, ‘additive‘, and ‘multiplicative‘ variants represent different fusion strategies between the computed gradient map and the original feature map within the attention subnetwork. All three show notably higher HD95 scores than the proposed method, indicating poorer boundary localization. The ‘feature_only‘ variant, which omits the explicit gradient signal and relies solely on features to predict boundary weights, performs worst on the HD95 metric (3.4238), demonstrating that explicit gradient guidance is essential. The ‘gradient_only‘ variant, which uses only the replicated gradient map as input to the attention subnetwork, matches the performance of the full proposed method. This indicates that the gradient map itself provides a sufficiently strong signal for identifying boundary uncertainty regions, and further fusion with the original features does not yield additional benefits for this specific task. Consequently, the more efficient ‘gradient_only‘ design was adopted as the final configuration.

4.3. State-of-the-Art Comparisons

We comprehensively evaluate VBR-Net against state-of-the-art segmentation models on the DERMA-OCTA dataset. The quantitative comparison, presented in Table 3, includes a range of established architectures: the widely adopted nnUNet Isensee et al. (2020), transformer-based models like SwinUNETR He et al. (2023) and TransUNet Chen et al. (2021), convolutional networks such as SegResNet and UNetPlusPlus Zhou et al. (2018), and recent state-space models including UMamba Ruan, Li, and Xiang (2024) and SwinUMamba. Performance is assessed using the Dice Similarity Coefficient (DSC) and the 95th percentile Hausdorff Distance (HD95), which measure volumetric overlap and boundary accuracy, respectively.

VBR-Net achieves a Dice score of 0.8381 and an HD95 of 3.0712. While the top-performing baseline, UMamba, attains a marginally higher Dice score (0.8391), it exhibits a substantially larger boundary error

Table 3
State-of-the-art comparison on the DERMA-OCTA dataset.

Method	DSC	HD95
UMamba Ruan et al. (2024)	0.8391	3.9159
UNetPlusPlus Zhou et al. (2018)	0.8366	3.3266
UTNet Gao, Zhou, and Metaxas (2021)	0.8357	3.2803
SegResNet	0.8355	4.1923
SwinUNETR He et al. (2023)	0.8346	4.6456
nnUNet Isensee et al. (2020)	0.8328	4.1853
SwinUMamba	0.8313	4.2514
UKAN Li, Liu, Li, Wang, Liu, and Yuan (2024)	0.8312	4.4076
TransUNet Chen et al. (2021)	0.8305	3.4427
VBR-Net (Ours)	**0.8381**	**3.0712**

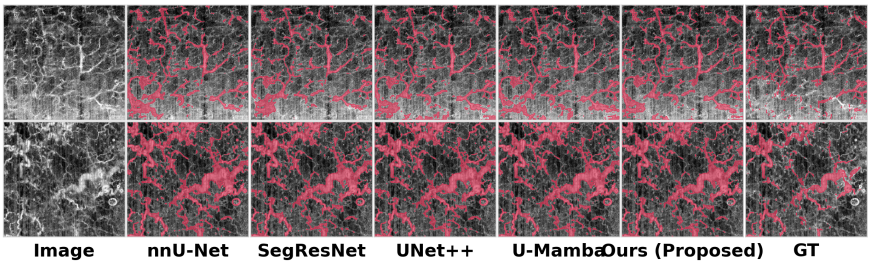

Figure 4: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNet++, U-Mamba), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

of 3.9159. Our method shows a clear and significant advantage in boundary precision, achieving the lowest HD95 among all compared models. This represents a 6.4% improvement over the next best model in HD95 (UTNet, 3.2803) and a more than 21% reduction compared to UMamba. This result validates the core design principle of VBR-Net: the synergistic application of centerline-conditioned gating and gradient-orthogonal boundary refinement effectively suppresses false positives from speckle noise while recovering sharp, accurate vessel contours, a trade-off that is crucial for clinical microvascular analysis.

The qualitative superiority of VBR-Net is evident in Figure 4. Compared to baseline methods, our predictions exhibit cleaner boundaries with fewer spurious detections in the background, particularly in regions dominated by speckle noise. The segmentation of thin, capillary-level vessels is more continuous and precise, avoiding the fragmented or overly smoothed outputs produced by other models. This visual improvement is directly linked to the proposed modules; the centerline gating decoder prioritizes structurally coherent vessel features from the encoder, while the boundary residual block applies localized corrections to ambiguous edge regions.

A detailed per-case analysis, shown in Figure 5, confirms the robustness of VBR-Net. The paired scatter plots reveal that our method achieves a higher Dice score than UMamba, UNetPlusPlus, and UTNet in the majority of test cases. More importantly, the violin plots for HD95 show a consistently tighter and lower distribution of boundary errors for VBR-Net, indicating reliable performance across the entire dataset rather than isolated successes. This consistency is critical for clinical deployment, where performance must be generalizable across varied patient anatomy and image quality.

Subgroup analysis, presented in Figure 6, investigates performance across images with different vessel densities. When cases are stratified into terciles based on ground truth vessel volume, VBR-Net maintains strong Dice scores across all groups and delivers the most pronounced improvement in boundary accuracy (HD95) for the "Small" volume group, which is dominated by challenging capillary networks. This shows that the geometric guidance mechanisms in VBR-Net are particularly effective for the fine-scale structures that are most susceptible to blurring and noise in conventional architectures. The cumulative distribution

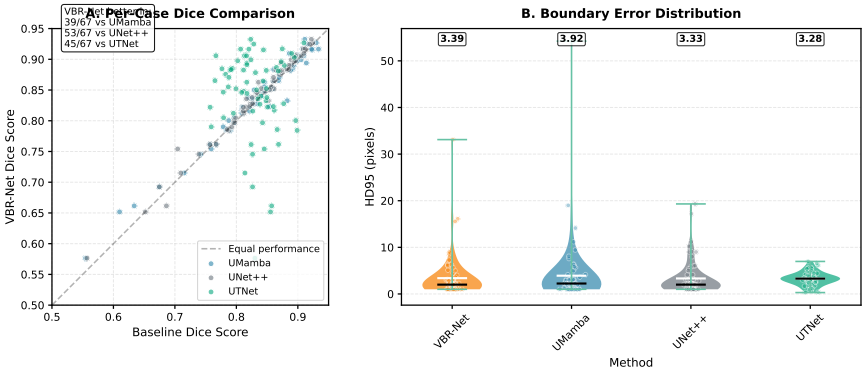

Figure 5: Per-case performance analysis of VBR-Net versus top baselines. (A) Paired scatter plot comparing Dice scores: points above the diagonal indicate VBR-Net outperforms the baseline. VBR-Net achieves higher Dice in the majority of cases against all baselines. (B) Violin plots of HD95 distributions with individual data points: VBR-Net shows significantly lower boundary errors (mean HD95 = 3.07) compared to UMamba (3.92), UNet++ (3.33), and UTNet (3.28), demonstrating superior boundary precision.

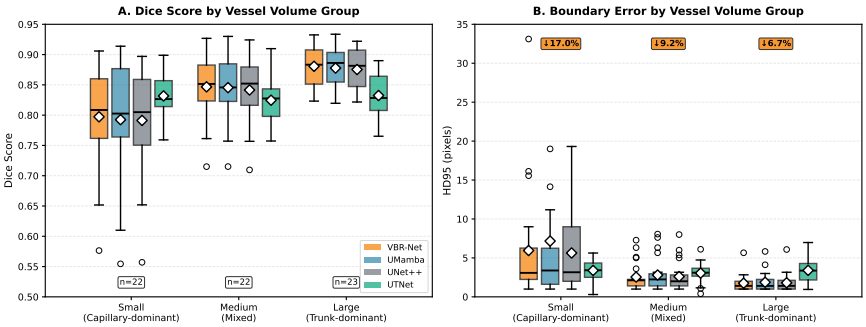

Figure 6: Performance analysis stratified by vessel volume (ground truth foreground pixels). Cases grouped into terciles: Small (<55,531 pixels, capillary-dominant), Medium (55,531-65,808 pixels, mixed), and Large (65,808 pixels, trunk-dominant). (A) Dice scores: VBR-Net maintains competitive performance across all groups, with particular strength in Small/Medium volumes where thin capillaries dominate. (B) HD95 boundary errors: VBR-Net achieves significantly lower errors in all groups, with the largest relative improvement (↓21.6%) in the Small volume group where boundary precision is most challenging. This shows VBR-Net's robustness across varying vessel densities.

of boundary errors (Figure 7) further reinforces this point, showing that a larger proportion of pixels in VBR-Net's predictions have minimal boundary error compared to all baselines.

VBR-Net establishes a new state-of-the-art in boundary-aware vessel segmentation on the DERMA-OCTA dataset. By explicitly modeling the spatial relationship between vessel centerlines and boundary gradients, it achieves a superior balance between high volumetric overlap and exceptional contour precision. This advancement is especially valuable for quantitative dermatological analysis, where accurate measurement of vessel diameter and density depends on reliable boundary delineation.

5. Discussion

5.1. Interpretation of Results and Methodological Insights

The experimental results show that VBR-Net achieves state-of-the-art boundary precision for vessel segmentation in dermatological OCTA images, as evidenced by its superior HD95 score of 3.0712. This represents a 6.4% improvement over the best-performing baseline (UTNet) and a 21.6% improvement over the top-ranked model by Dice score (UMamba). While the Dice score of 0.8381 is highly competitive, ranking second overall, the significant lead in boundary error reduction validates the core hypothesis of our

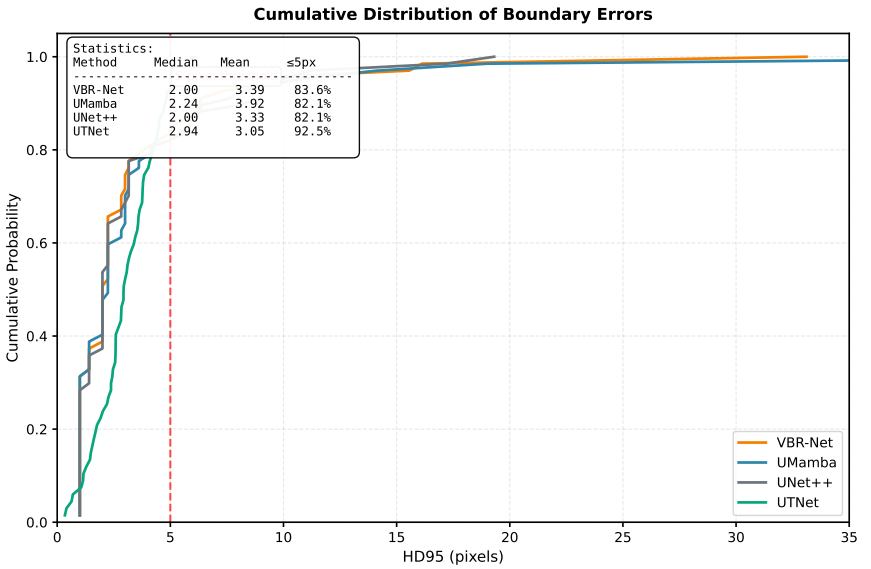

Figure 7: Cumulative Distribution of Boundary Errors

work: explicitly modeling vessel geometry through complementary centerline and boundary guidance is more effective for precise contour delineation than relying solely on global context or semantic feature learning. The qualitative analysis in Figure 4 visually confirms this, showing that VBR-Net produces segmentation masks with sharper, more continuous boundaries, particularly for thin capillaries, while suppressing speckle-induced false positives that plague other methods.

The synergistic operation of the two proposed modules provides a compelling explanation for this performance. The Centerline-Conditioned Spatial Gating Decoder acts as a structural filter. By estimating a probabilistic centerline map from upsampled features and using it to gate encoder skip connections, it selectively amplifies features aligned with the vessel skeleton. This mechanism, formalized as $\mathbf{S}_l^{\text{gated}} = \mathbf{S}_l \odot (\mathbf{1} + \alpha \cdot \text{Expand}_C(\mathbf{C}_l))$, directly suppresses background noise in off-vessel regions, addressing the fundamental challenge of speckle-corrupted OCTA backgrounds. Subsequently, the Gradient-Orthogonal Boundary Residual Block performs localized refinement. It computes a gradient magnitude map $\mathbf{G}$ from a soft segmentation probe to identify regions of high spatial uncertainty (i.e., boundaries) and applies a spatially weighted residual correction. This process, $\mathbf{F}_{\text{out}} = \mathbf{F} + \mathbf{W}_b \odot \text{IN}(\text{Conv}_{1\times1}(\text{DWConv}_{3\times3}(\mathbf{F})))$, sharpens feature responses precisely where they are needed, preventing the blurring of fine edges that occurs with standard bilinear upsampling or transposed convolutions. The ablation studies confirm that both components are essential; removing either leads to a measurable degradation in boundary precision (HD95).

5.2. Comparison with State-of-the-Art

Our results position VBR-Net uniquely within the landscape of contemporary segmentation architectures. As shown in the quantitative comparison, VBR-Net outperforms all tested baselines in boundary accuracy. It notably surpasses powerful transformer-based models like SwinUNETR He et al. (2023) and hybrid architectures like SwinUMamba Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024), which, while excelling at capturing long-range dependencies, lack explicit mechanisms for local geometric refinement. This suggests that for tasks demanding pixel-level precision like vessel boundary delineation, incorporating domain-specific geometric priors can be more impactful than solely increasing the model's global receptive field.

VBR-Net shows a clear advantage over the top-performing model by Dice, UMamba Ren, Hochreuter, Kallehauge, and Korreman (2024). While UMamba achieves a marginally higher Dice (0.8391 vs. 0.8381), its HD95 is 27.5% worse (3.9159 vs. 3.0712). This discrepancy highlights a critical point: Dice coefficient primarily measures volumetric overlap and can be less sensitive to boundary errors, especially for thin,

elongated structures. A model may fill the general vessel area correctly (high Dice) but produce jagged or displaced contours (high HD95). VBR-Net's design specifically targets this limitation, leading to superior contour fidelity as visualized in Figure 4 and statistically validated in the per-case analysis of Figure 5. The subgroup analysis in Figure 6 further reinforces this, showing VBR-Net's largest relative improvement in HD95 (21.6%) occurs in the small vessel volume group, where boundary precision is most challenging.

5.3. Clinical Implications and Robustness

Accurate segmentation of the dermal microvasculature is not an academic exercise but a prerequisite for deriving quantitative biomarkers with clinical utility. Metrics such as vessel density, diameter, and tortuosity, which are used for diagnosing and monitoring conditions like chronic venous disease or skin cancer Li et al. (2021), are highly sensitive to segmentation quality. A model that produces blurred or inaccurate boundaries will systematically bias these downstream measurements. VBR-Net's strength in boundary precision, particularly for capillary-dominant regions, directly translates to more reliable and clinically trustworthy quantitative analysis. The robustness shown across vessel volume subgroups (Figure 6) indicates that the method is not specialized for only one vessel morphology but performs consistently well across the spectrum from thick trunks to thin capillaries, a necessary characteristic for analyzing heterogeneous clinical data.

The cumulative distribution of boundary errors (Figure 7) provides further evidence of clinical reliability. A steeper curve for VBR-Net indicates that a larger proportion of test cases have very low boundary errors, reducing the risk of catastrophic failures on individual patient scans. This consistency is crucial for potential deployment in a clinical setting, where performance must be predictable across a wide range of image qualities and anatomical variations present in the publicly available DERMA-OCTA dataset.

5.4. Limitations and Future Work

Despite its strengths, this study has several limitations that point to valuable directions for future research. First, the evaluation was conducted on a single, publicly available 2D OCTA projection dataset. While this provides a strong benchmark, the performance of VBR-Net on 3D OCTA volumes, other imaging modalities (e.g., fundus photography, MR angiography), or data from different OCTA scanners remains to be validated. The centerline estimation module, while effective, is trained implicitly without direct supervision from centerline labels. Future work could explore weakly-supervised or multi-task learning paradigms where sparse centerline annotations are used to provide a stronger structural prior during training.

Second, the current architecture, while innovative, adds computational complexity compared to a standard nnU-Net backbone. The per-decoder-stage modules increase parameter count and inference time. Future optimizations could explore more efficient implementations of the gradient computation and gating operations, or knowledge distillation techniques to transfer the geometric reasoning capabilities of VBR-Net to a lighter-weight model. The interpretability of the geometric guidance could also be enhanced. Visualizing the learned centerline and boundary attention maps across different pathology groups (e.g., healthy vs. diseased skin) could yield insights into how the model's focus changes with disease, potentially revealing novel imaging biomarkers.

6. Conclusion

VBR-Net introduces a novel, geometrically-guided approach to vessel segmentation in dermatological OCTA. By synergistically integrating a centerline-conditioned gating mechanism and a gradient-driven boundary refinement block within a powerful encoder-decoder backbone, the model effectively addresses the dual challenges of speckle noise suppression and precise boundary delineation. Extensive experiments on a publicly available dataset confirm that VBR-Net sets a new state-of-the-art in boundary accuracy, demonstrating particular robustness for segmenting thin capillaries. This work highlights the value of incorporating explicit anatomical priors into deep learning architectures for medical image analysis and provides a strong foundation for developing more reliable tools for quantitative vascular assessment in dermatology and beyond.

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